Creative assignment – Models of Mushrooms of the Earth

Materials: colored and wheat-flour modeling dough, used hair oil capsules, toothpick, used plastic bowl, note papers, masking tape

Fig. 1 materials which I used (colored dough, tape, capsules)

Artwork Description

: I made models of diverse mushrooms of the world with colored dough and some discarded materials. I depicted a young wrinkled pink mushroom (*Rhodotus palmatus)*, indigo milk cap (*Lactarius indigo*) and jack-o'-lantern mushroom (*Omphalotus olearius*). Each mushroom model and its substrate have distinct specific colors. *Rhodotus palmatus* has a pink cap and reddish droplets that appear on its fruiting body. I tried to illustrate such figures with colored dough and pink ellips hair oil capsules. Some of the capsules melted so that they could look like the liquid of the mushroom. Also, the *Rhodotus* is on the dark brown rotting trees. *Lactarius indigo* is upside down so that it can show its gills structure, which are running down the length of the stem. I intended to express the light blue cap whose margin is rolled inward and the gills darker than the cap. *Omphalotus olearius* forms a colony on buried roots in tropical forests. I tried to represent the attachment to cap to base of stem. The actual color of *O. olearius* is bright orange, but I chose yellow in order to include its bioluminescent properties. Each mushroom model accompanied by a cloud-shaped label detailing the specific mushroom, which includes the color, substrate, order-family name and structures.

Educational Aspects

: My work indicates the diverse form of basidiomycota all over the world like Europe, China and North America. Viewers can identify not only the individual features of each mushroom but also the whole properties of fungi (Basidiomycota). For example, there are figures of caps and stipe, milky latex and the structure of gills. By observing the structure of the gills, viewers can also understand the mechanism of spore dispersal, which is essential for fungal reproduction. Also, the models allow viewers to visually compare the substrate environments and growth patterns of each mushroom.

Fig. 2 The picture of the final work

*The journal content I referred to

The wrinkled peach mushroom(young) (Agaricales, Physalacriaceae)

: One such example is the wrinkled peach mushroom, *Rhodotus palmatus* (Bull.) Maire. (5,6) This unmistakable mushroom, with its striking pink caps, deep engravings, and blood-like drops covering younger fruiting bodies, is a pioneer fungus on freshly rotting hardwoods, with a preference for elm trees.

In nature, Rhodotus palmatus is sometimes seen "bleeding" a red- or orange-colored liquid. A similar phenomenon has also been observed when it is grown in laboratory culture on a petri dish: the orange-colored drops that appear on the mat formed by fungal mycelia precede the initial appearance of fruit bodies.

Indigo milk cap (Russulales, Russulaceae)

: A beautiful, blue mushroom species that exudes a milky blue latex when cut with a knife or bruised, that slowly turns green as it is exposed to air.

Lactarius indigo is not common, but is widespread in its distribution, growing naturally in eastern North America, East Asia, and Central America; it has also been reported in southern France.

L. indigo grows on the ground in both deciduous and coniferous forests, where it forms symbiotic relationships - called mycorrhizal associations - with the roots of plants. The cap has a diameter of two to six inches. The margin of the cap is rolled inwards when young but unrolls as it matures. The cap surface is indigo blue when fresh but fades to a paler grayish or ethereal silvery-blue. Sometimes it's adorned with greenish splotches. It is often zonate: marked with concentric lines that form alternating pale and darker zones, like the layered ruffles of many a dress, and the cap may have dark blue spots, especially towards the edge. Young caps are sticky to the touch.

The gills of the mushroom range from adnate (squarely attached to the stem) to slightly decurrent (running down the length of the stem), and crowded close together. Their color is an indigo blue, becoming paler with age or staining green with damage.

jack-o'-lantern mushroom (Agaricales, Omphalotaceae)

: The *O. olearius* is a type of toxic mushrooms, bright orange in color, that lacks the fruity taste, and glows in the dark (Aoki et al., 2020). The *O. olearius* grows mainly in tropical and subtropical forests, appearing in the USA, Australia, China, Tanzania etc. Some incidents of mistakenly consuming *O. olearius* were gradually appearing in the reports, simply because the *O. olearius* is remarkably identical in appearance to the C. albovenosus and has overlapping habitats (Bal et al., 2016). The C. albovenosus, an ectomycorrhizal fungus classified under the Basidiomycetes phylum, often grows as a mycorrhiza relationship with pines, oaks, other broadleaves, and conifers a member of Cantharellales (Chen & Xu, 2024).

The whole mushroom does not glow—only the gills do so.

Reference

1. Sandargo, B., Michehl, M., Stadler, M., & Surup, F. (2020). Antifungal sesquiterpenoids,

rhodocoranes, from submerged cultures of the wrinkled peach mushroom, *Rhodotus palmatus*. *Journal of Natural Products*, *83*(3), 720–724. https://doi.org/10.1021/acs.jnatprod.9b00871

2. Miller OK Jr; Palmer JG; Gillman LS. (1980). "The fruiting and development of *Rhodotus palmatus* in culture". *Mycotaxon*. **11** (2): 409–19.

3. Sargent, C. (2021, October 25). *Species of the week:* Lactarius indigo. One Earth. https://www.oneearth.org/species-of-the-week-lactarius-indigo/

4. Yao, J., Zhu, J., Marchioni, E., Zhao, M., Li, H., & Zhou, L. (2025). Discrimination of wild edible and poisonous fungi with similar appearance (*Cantharellus albovenosus* vs *Omphalotus olearius*) based on lipidomics using UHPLC-HR-AM/MS/MS. *Food Chemistry*, *466*, Article 142189.

https://doi.org/10.1016/j.foodchem.2024.142189